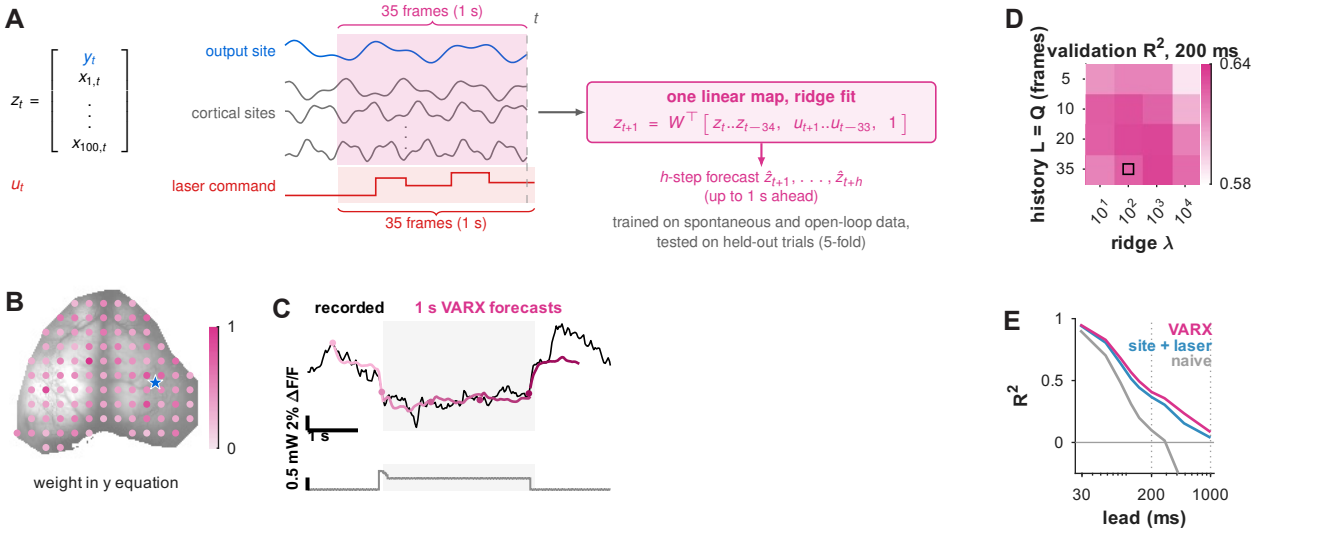

A
35 frames (1 s)
t
output site
cortical sites
laser command
35 frames (1 s)
one linear map, ridge fit
h-step forecast
(up to 1 s ahead)
trained on spontaneous and open-loop data,
tested on held-out trials (5-fold)
D
validation R², 200 ms
history L = Q (frames)
ridge λ
0.64
0.58
B
weight in y equation
C
recorded
1 s VARX forecasts
1 s
0.5 mW 2% ΔF/F
E
VARX
site + laser
naive
lead (ms)